OCEANVERSE AI

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ABSTRACT

OceanVerse AI is an AI-powered platform that integrates ocean monitoring, biodiversity analysis, environmental analytics, and real-time safety. It analyzes ocean parameters such as temperature, salinity, waves, currents, pollution, weather, and cyclone risk to predict marine risks and identify **Safe, Caution, and High-Risk zones**. The platform provides interactive dashboards, maps, alerts, biodiversity monitoring, and safer route recommendations, supporting **marine conservation, climate awareness, sustainable ocean activities, SDG 13 and SDG 14**.

MOTIVATION

OceanVerse AI is motivated by the need to address growing ocean challenges such as pollution, climate change, biodiversity loss, and extreme weather by combining scattered ocean data into one unified platform. It uses AI to provide meaningful predictions, real-time safety alerts, and risk-zone identification, helping fishermen, travelers, researchers, and coastal communities make safer and more informed decisions while supporting marine conservation and sustainable ocean activities.

Literature Review – Papers 1–3

Paper Name	Year	Methodology	Outcome	Evaluation Measure (Accuracy)	Limitations
A Survey on Usage of Ontology in Different Domains	2012	Survey of ontology applications using semantic representation, controlled vocabulary, relationships, and interoperability; includes the marine domain.	Shows that ontology supports common semantics, knowledge sharing, reuse, and interoperability between applications.	Not applicable – no evaluation was attempted.	Broad multi-domain survey rather than an ocean-specific platform; marine ontology coverage is limited.
JMarine: A Marine Biodiversity Database for Johor	2013	Prototype web-based biodiversity database using a unified marine biodiversity data model, Drupal/PHP/MySQL, and species, spatial, taxonomy, bathymetry, and mapping data.	Created a web-accessible marine biodiversity knowledge portal for storing and sharing biodiversity information.	Not reported – database/platform study, not an ML classification problem.	Regional focus on Johor; mainly biodiversity information management rather than AI prediction.
Identifying the Top Issues of Marine Invasive Alien Species in Europe	2014	Review/discussion of marine invasive alien species, examining prevention, taxonomy, data gaps, terminology, validation, and pathways.	Identified major issues and emphasized standardized and scientifically validated marine biodiversity information systems such as AquaNIS and EASIN.	Not reported.	Primarily policy/management oriented; no AI model or quantitative prediction system.

Literature Review – Papers 4–6

Paper Name	Year	Methodology	Outcome	Evaluation Measure (Accuracy)	Limitations
Biodiversity Baseline for Large Marine Ecosystems: An Example from the Barents Sea	2015	Ten-year demersal and pelagic fish-trawl dataset with hierarchical partitioning of biodiversity and spatial/temporal analysis.	Produced a spatio-temporal biodiversity baseline for detecting ecosystem change and comparing future observations.	Not applicable – biodiversity/diversity statistics rather than classification accuracy.	Measures are sensitive to sampling effort and aggregation; some statistical approaches are difficult for biomass/coverage data.
Marine Anoxia and Delayed Earth System Recovery After the End-Permian Extinction	2016	Paired uranium concentration and $\delta^{238}U$ isotope records with uranium box modelling and statistical analysis to reconstruct marine redox conditions.	Identified increased seafloor anoxia and linked prolonged shallow-ocean anoxia with delayed biodiversity and ecosystem recovery.	Not applicable – model/statistical reconstruction rather than AI classification.	Depends on assumptions about carbonate records and uranium flux/model parameters.
ACCESS Climate Data Management	2017	Metadata-driven scientific data-management system using metadata checking, dataset discovery, download, visualization, and transformation; METAMOD and THREDDS.	Improved sharing, discovery, preservation, and interoperability of climate/ocean scientific datasets.	Not applicable – functional/data-management evaluation.	Incomplete integration between climate and non-climate repositories; limited scientific user uptake.

Literature Review – Papers 7–9

Paper Name	Year	Methodology	Outcome	Evaluation Measure (Accuracy)	Limitations
Large-scale Ocean Connectivity and Planktonic Body Size	2018	Malaspina 2010 biological observations combined with global circulation-model transit times; β-diversity, time-decay analysis, hierarchical clustering, and network analysis	Demonstrated that ocean-current dispersal, local abundance, and body size influence global planktonic and micro-nektonic biodiversity patterns	$R^2 = 0.93$ for body-size/local-abundance relationship; not an ML accuracy measure.	Mainly focuses on plankton/micro-nekton connectivity; not a unified platform or AI system.
A Framework for the Development of a Global Standardised Marine Taxon Reference Image Database (SMarTaR-ID) to Support Image-Based Analyses	2019	Standardized marine taxon image database; Darwin Core mapping; expert curation and reference image catalogues.	Provides a common framework for combining marine image datasets and supporting future AI-based taxon identification.	Not reported – framework/database proposal.	Image identification still requires ground-truthing with specimens and expert taxonomic validation.
Mapping Human Pressures on Biodiversity Across the Planet Uncovers Anthropogenic Threat Complexes	2020	Global spatial datasets representing climate change, human use, pollution, and invasive-species pressure; standardized spatial analysis and clustering.	Identified spatial combinations of anthropogenic pressures affecting biodiversity and compared terrestrial and marine systems.	Not reported – spatial relationship/clustering analysis.	Depends on availability, resolution, and compatibility of global datasets; relationships do not directly establish biodiversity causation.

Literature Review – Papers 10–12

Paper Name	Year	Methodology	Outcome	Evaluation Measure (Accuracy)	Limitations
Autonomous Underwater Monitoring System for Detecting Life on the Seabed by Means of Computer Vision Cloud Services	2020	AUV + IoT gateway + cloud/edge computing + CNN-based underwater object detection; autonomous tracking and georeferencing of Pinna nobilis.	Demonstrated autonomous detection, tracking, and georeferencing using a hybrid cloud/edge architecture; edge computing reduced latency.	83.8% accuracy for the edge model at 2.13 m.	Underwater image-quality and communication limitations; focuses on one target species and requires cloud/edge architecture.
Coral Reef Mapping with Remote Sensing and Machine Learning: A Nurture and Nature Analysis in Marine Protected Areas	2021	WorldView-3 + Landsat-8; GWR bathymetry; SVM habitat classification; MaxEnt species-distribution modelling; human-use mapping.	Generated a 15-class coral reef habitat map and identified environmental/human factors affecting Millepora alcicornis distribution.	79% overall accuracy; SVM cross-validation = 0.91; Kappa = 0.79.	Regional study; dependent on image quality, water transparency, bathymetry, and field data; not a unified multi-source platform.
Smart Oceans: Artificial Intelligence and Marine Protected Area Governance	2022	12 expert interviews; review of 50 dynamic marine-management papers and 500 bioacoustics/ecoacoustics papers; examination of digital marine forecasting and AI-enabled MMPA case studies.	Demonstrated potential for combining remote sensing, acoustic monitoring, sensors, bio-logging, AIS, oceanographic models, and AI for dynamic marine protected areas.	Not reported – qualitative/case-study evaluation.	Data ownership, robustness, privacy, energy/e-waste, digital inequality, and unequal access to Smart Ocean technologies.

Literature Review – Papers 13–15

Paper Name	Year	Methodology	Outcome	Evaluation Measure (Accuracy)	Limitations
Accelerating Ocean Species Discovery and Laying the Foundations for the Future of Marine Biodiversity Research and Monitoring	2023	Ocean Census framework integrating expeditions, taxonomy, DNA barcoding, imaging, databases, cloud/SaaS processing, data exchange, and standardization.	Proposes a global cyber-biodiversity system with standardized processing, interoperability, and exchange with OBIS, WoRMS, WOD, and	Not reported – framework/policy review.	Many ocean regions remain poorly sampled; data-management standards differ across data types, especially imagery; regional capacity varies.
High Seas in the Cloud: The Role of Big Data and Artificial Intelligence in Support of High Seas Governance – The Sargasso Sea Pilot	2024	Big Data + AI + remote sensing + autonomous observations + data-science governance cycle; multi-source data integration.	Shows how AI and heterogeneous ocean observations can support high-seas biodiversity assessment, monitoring, and governance.	Not reported – policy/practice review.	Heterogeneous data repositories, difficult data access, interoperability, and need for automated aggregation/analysis workflows.
Towards Standardizing Automated Image Analysis with Artificial Intelligence for Biodiversity	2024	Review/opinion of AI-based marine image analysis; deep learning, automated detection/recognition/tracking; recommendations for data, model, and code standards.	Recommends standardized AI workflows and a user-friendly AI-cloud platform for automated biodiversity image analysis.	Fish4Knowledge studies cited 99.45% and 98.79% recognition accuracy; these are cited studies, not accuracy of this review.	Blurry/low-light underwater images; difficult species-level identification; performance varies with dataset, architecture, and data division.

Literature Review – Papers 16-17

Paper Name	Year	Methodology	Outcome	Evaluation Measure (Accuracy)	Limitations
Recent Advancements and Challenges in Artificial Intelligence for Digital Twins of the Ocean	2025	Review of ML, DL, computer vision, hybrid physics/ML models, and Digital Twins of the Ocean; satellite + in-situ + numerical-model data integration.	Shows potential for real-time forecasting, water-quality monitoring, biodiversity monitoring, pollution detection, and decision support through AI-enabled ocean digital	No single accuracy value – review synthesizes results from multiple models.	Data scarcity/quality, interoperability, regional model transfer, computational requirements, black-box models, and uncertainty.
Technologies for Marine Biodiversity Monitoring and Mapping: A Systematic Review	2026	PRISMA/JBI systematic review of marine biodiversity monitoring technologies; analysis of AI/ML, robotics, sensors, AUVs, ROVs, IoUT, EOVs, and monitoring platforms.	Shows increasing use of AI, autonomous systems, and real-time technologies while identifying gaps in interoperability, standardization, scalability, and long-term monitoring.	Not applicable – systematic review; no single AI accuracy value.	Heterogeneous data formats, interoperability problems, insufficient AI-performance reporting, cost, scalability, limited autonomy, and lack of standardized metrics.

EXISTING

- Existing systems mainly focus on specific ocean or biodiversity tasks.
- Ocean data is scattered across different platforms and sources.
- Current systems face challenges in data integration, interoperability, and real-time processing.
- AI tools are often limited to specific regions, species, or applications.
- A unified platform combining ocean data, biodiversity, AI, and visualization is needed.

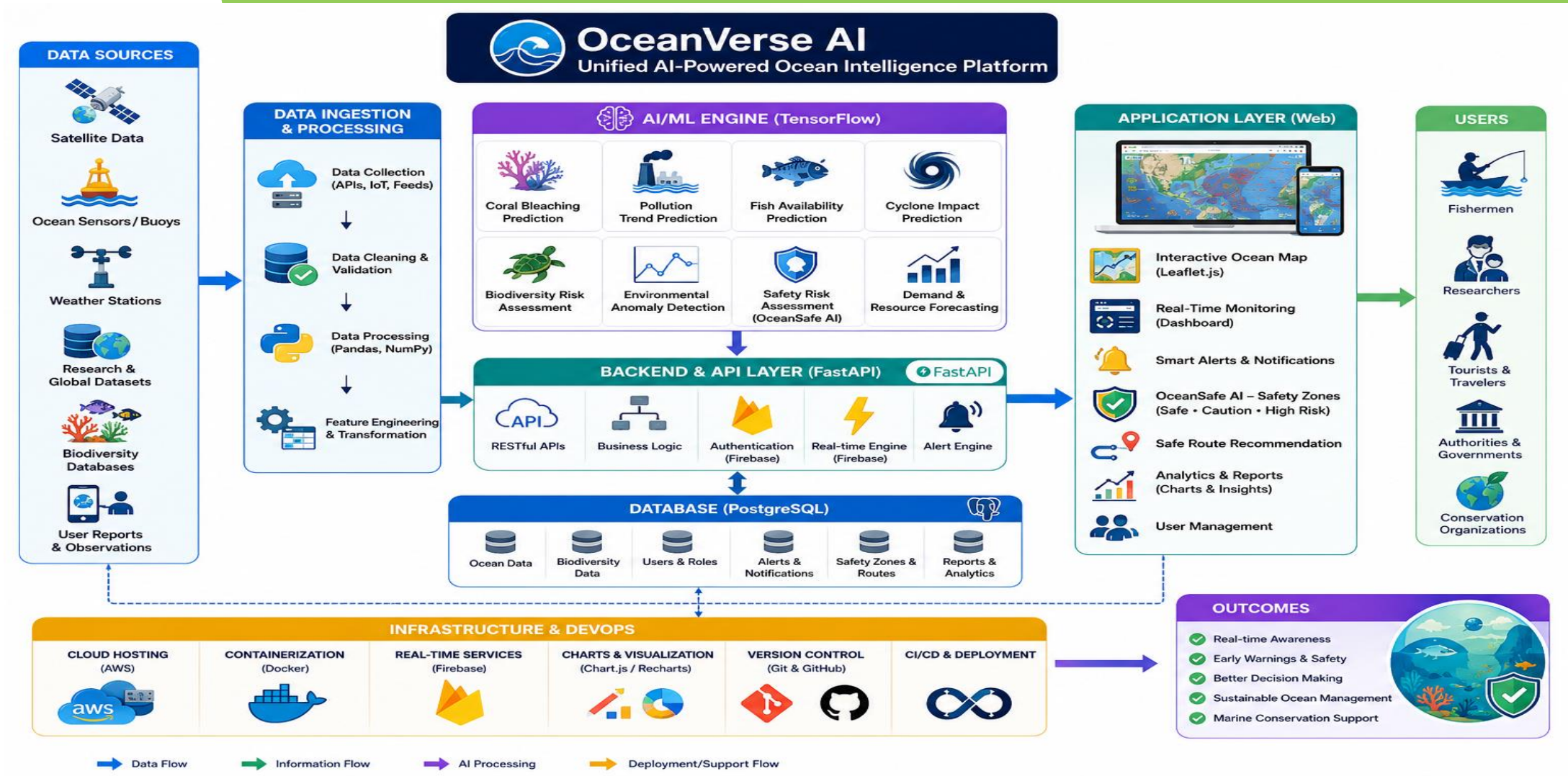
PROPOSED SOLUTION

- Develop a unified AI-powered platform for ocean monitoring, biodiversity, prediction, and safety.
- Integrate real-time ocean, weather, pollution, and environmental data in one system.
- Use AI to predict coral bleaching, pollution, fish availability, biodiversity risks, and cyclone impacts.
- Provide an interactive map showing ocean conditions, biodiversity, pollution, and safety zones.
- Identify Safe, Caution, and High-Risk zones with real-time alerts.
- Recommend safer routes and nearby safe zones for ocean users.
- Monitor marine species, coral reefs, mangroves, seagrass, and ecosystem health.
- Provide analytics, trends, insights, and downloadable reports.
- Support marine conservation and sustainable ocean management through data-driven decisions.

TOOLS & TECHNOLOGIES

Category	Technology
UI Development	Lovable, React.js
Styling	Tailwind CSS
Backend	FastAPI, Python
AI/ML	TensorFlow
Data Processing	Pandas, NumPy
Database	PostgreSQL
Maps	Leaflet.js
Cloud	AWS
Authentication	Firebase
Containerization	Docker
Charts	Chart.js / Recharts
Version Control	Git & GitHub

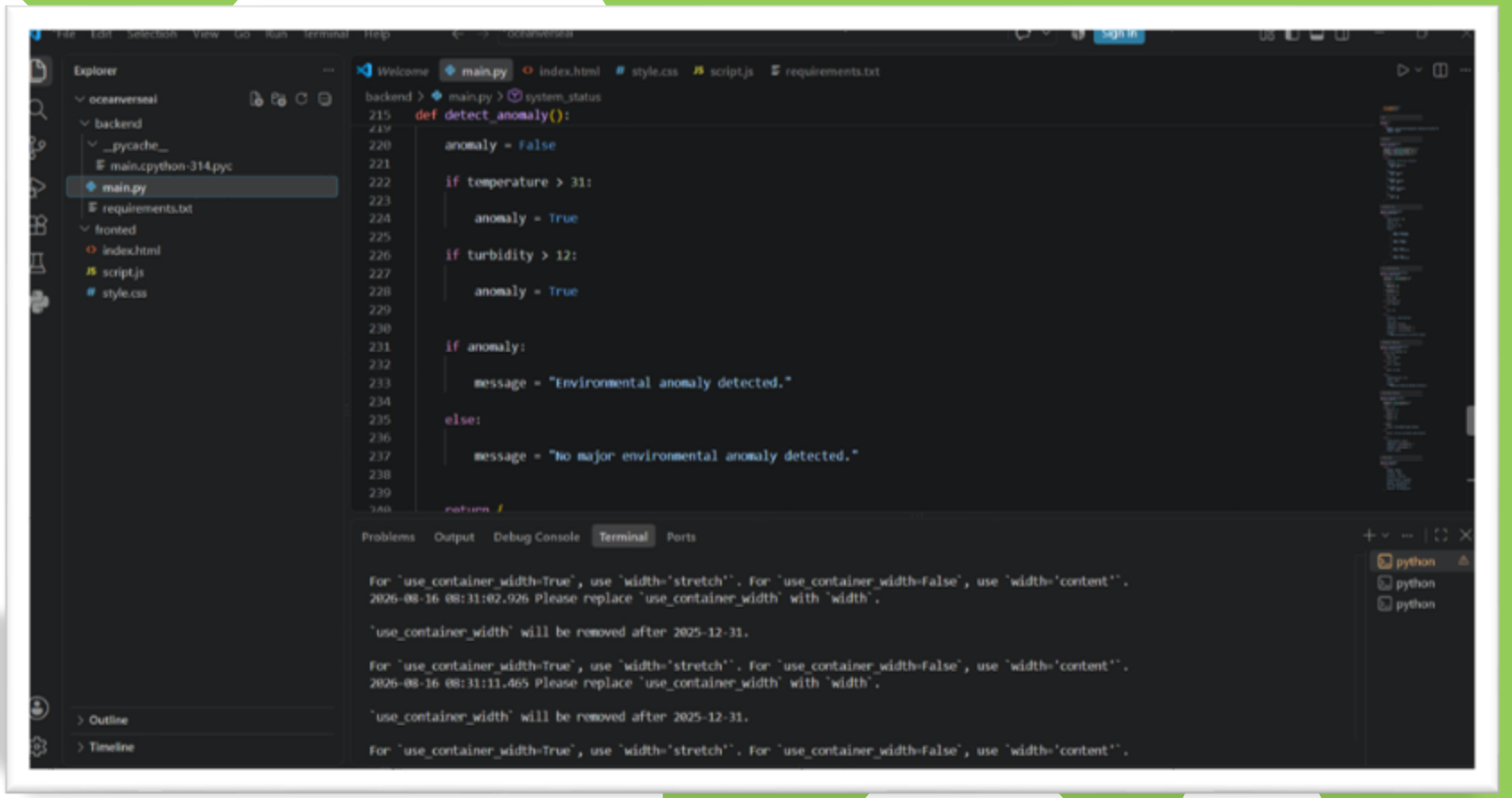
ARCHITECTURE DIAGRAM



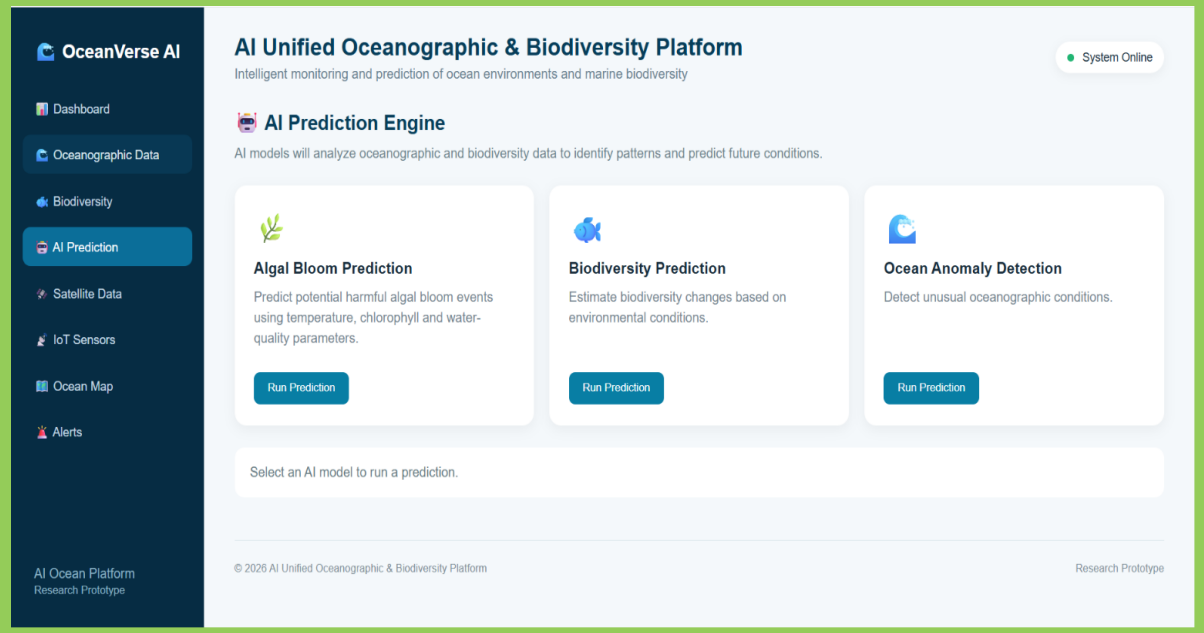
MODULE

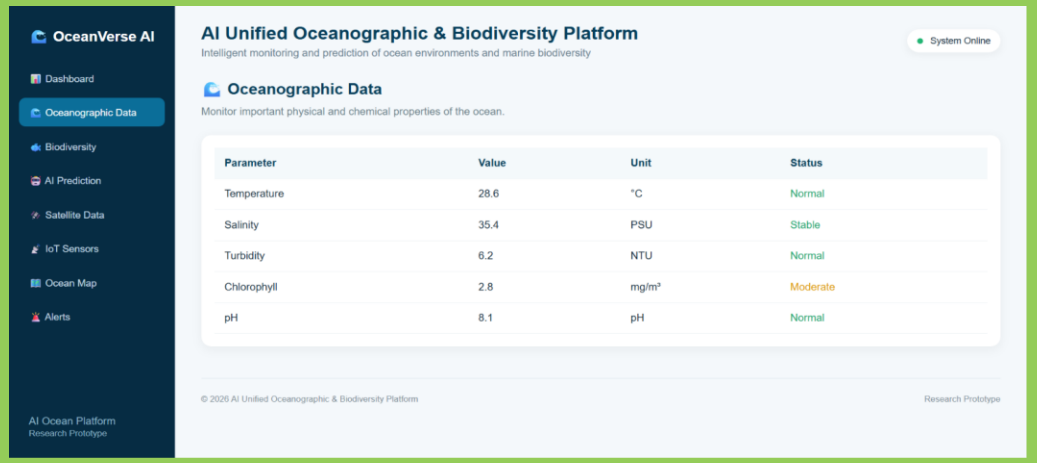
No.	Module	Main purpose
1	Unified Dashboard	Overall ocean and biodiversity status, charts, KPIs
2	Oceanographic Monitoring	Temperature, salinity, turbidity, pH, chlorophyll, etc.
3	Biodiversity Monitoring	Species, habitats, biodiversity indicators
4	AI Prediction & Analysis	Algal bloom prediction, anomaly detection, biodiversity prediction
5	Satellite Data Analysis	Satellite-derived ocean observations such as SST and chlorophyll
6	IoT Sensor Monitoring	ESP32/Arduino sensor readings and real-time monitoring
7	Interactive Ocean Map	Sensor locations, observations, biodiversity and environmental layers
8	Alerts & Reports	Environmental alerts, prediction warnings and downloadable reports

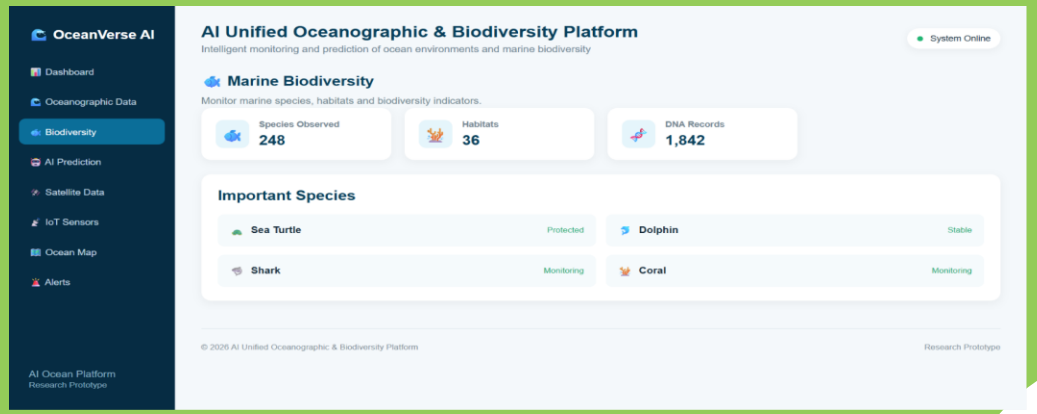
SCREENSHOT OF CODE

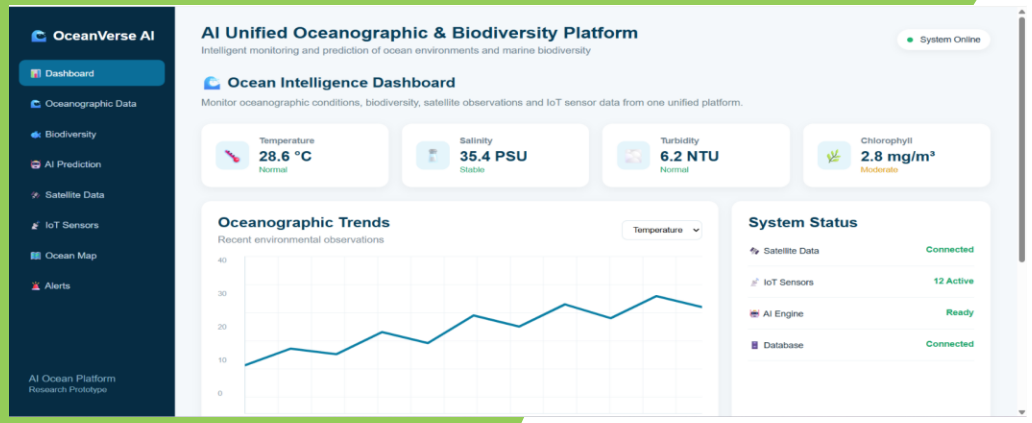


SCREENSHOTS OF OUR PLATFORM

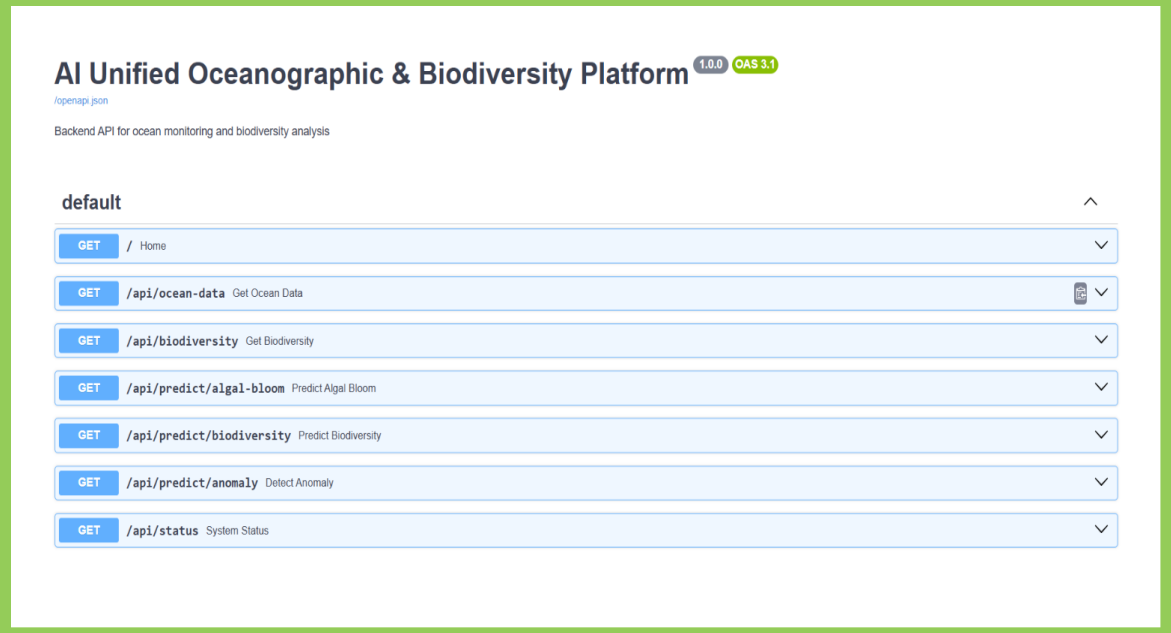


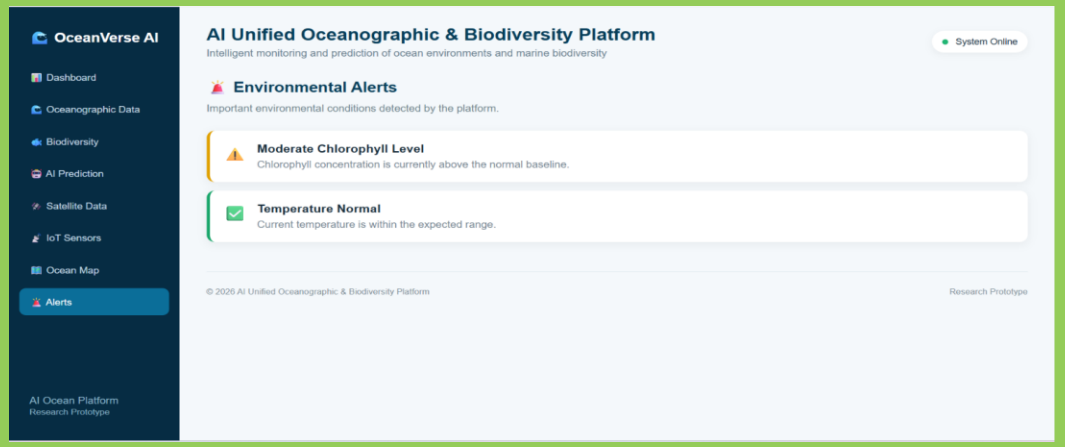


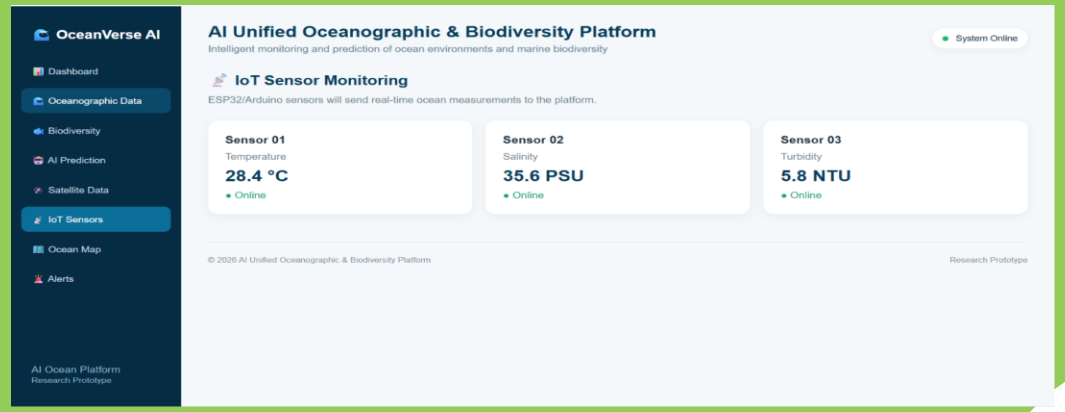


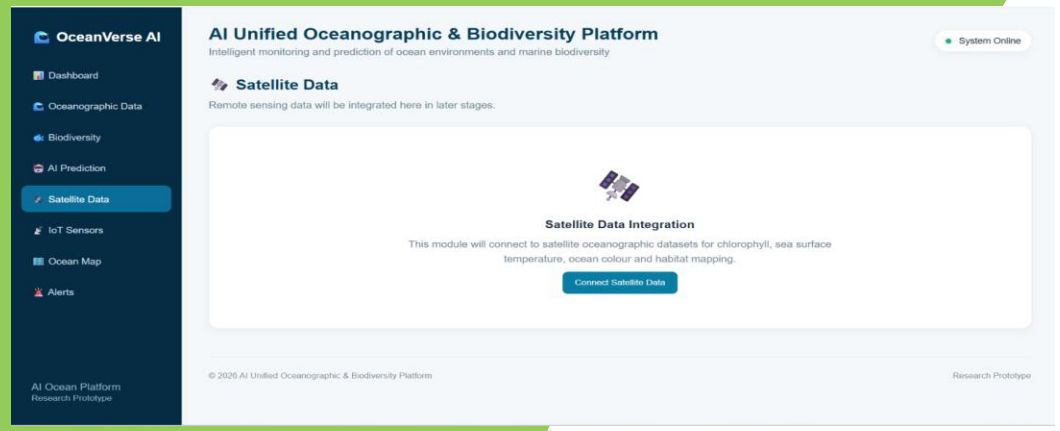


SCREEBSHOTS OF OUR PLATFORM









ACCURACY

Metric	Result
Accuracy	100%
Precision	100%
Recall	100%
F1-Score	100%
Model	Random Forest Classifier

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